

PV Abnormal Shading Detection Based on Convolutional Neural Network

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Abstract: Abnormal shading on the surface of photovoltaic panels will seriously damage its power generation efficiency and life. This paper established an abnormal shading detection system for photovoltaic panels. First, remove the photovoltaic panel and find the shading. Finally, the object detection algorithm is used to identify and classify the shading. Experimental results show that the system's abnormal shading classification accuracy is 94.5%, which is 23% higher than the old method.

Key Words: Photovoltaic, Shading, Convolutional Neural Network, Object Detection

1 Introduction

The penetration of photovoltaic (PV) technology has significantly increased worldwide in the last decade, with applications ranging from small rooftop or building-integrated PV systems to multi-MW PV power plants, at all kinds of installation locations, from isolated rural spaces to residential areas inside large cities. The variety of installation locations, especially in urban environment, often leads to operation at non-uniform illumination conditions because of the falling of abnormal object such as leaves and bird drops [1]. Fault detection of photovoltaic system plays an important role in its economic and sustainable development. These faults can be related to factors such as material interactions (corrosion of connectors, yellowing, browning of encapsulation material and discoloration of busbars) and environment factors such as soiling and shading. Soiling refers to the accumulation of snow, dirt, dust, leaves, pollen, and bird droppings on PV panels [2]. Shading may be a result of soiling or occurs due to obstructions caused by trees, buildings or chimneys. Thus, PV cells or modules may be partially or completely shaded during their operation. Shading is one of the most recurrent and damageable faults. In fact, this condition induces important degradation of PV system performances. Partial shading can lead to more than 20% of annual reduction in power production in residential applications as shown by [3]. Moreover, the presence of localized shading on PV modules leads to an overheating of the shaded cells despite the presence of bypass diodes. Using the infrared thermography (IRT) many studies prove the presence of hot spot zones on the shaded PV cells [4, 5]. Thus, the temperature increase in these zones leads to a thermal power dissipation [6], reduces considerably the PV module lifetime and

can damage the shaded cells [7]. An improved identification method based on sinusoidal echo state network (SES-N) is proposed to identify a class of periodic discrete-time dynamic nonlinear systems with or without noise [8]. Lei Liu et al. solved the problem of finite-time fault-tolerant control for a class of switched nonlinear systems in lower-triangular form under arbitrary switching signals [9]. Yao et al. proposed an adaptive neural network (NN) control approach for nonlinear pure-feedback systems with time-varying full state constraints [10]. Bo Zhao et al. proposed a model for optimizing the cleaning cycle of module dust and evaluating the cost for the PV power plants in China by the use of dust deposition monitoring with image recognition and two cleaning technologies [11]. It is beneficial to analyze the influence of foreign body shielding on the photovoltaic panel surface on its power generation efficiency, determine the cleaning cycle of the photovoltaic panel, determine the relationship between cleaning cost and power generation loss cost, and maximize the profit of photovoltaic power generation, which is conducive to the further development of the photovoltaic industry. Therefore, the abnormal shading detection of PV module is particularly significant. In recent years, a growing number of examples of artificial intelligence applied to photovoltaic fault detection have made significant achievement. The research on the detection of photovoltaic panel faults by infrared and EI images has developed rapidly, but the analysis of visible images is still the mainstream due to its convenience in obtaining. S, a et al. proposed a highly accurate power prediction with soiling location and classification model by combining weak supervised learning with convolutional neural network [12] [13] proposed deep-learning based detection of visible fault with use of aerial imaging for PV modules. Study [14] automatically detects visible fault in solar cells by using multispectral CNN. Nonetheless, more analysis focus on large-scale photovoltaic array images, while less research on abnormal shading detection on the surface of PV module. For small distributed photovoltaic

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plant can be recognize whether there is any abnormal shading keep out artificially, but for large grid-connected photovoltaic plants, artificial detection is not realistic. Therefore, in this paper, we proposed a PV abnormal shading detect system based on deep learning object detection algorithm YOLOv3 [15] and Mask R-CNN [16]. First, the PV module information in the image is extracted, and the abnormal shading of different shapes and sizes can be classified. The proposed abnormal shading detect system is divided into three steps. The first two steps use YOLOv3, and the object recognition part uses Mask R-CNN. This article first introduces the two algorithms, YOLOv3 and Mask R-CNN, and then introduces the system structure in detail. The experimental results and analysis are given in the end.

2 Method

In this subsection we have described briefly about the CNN models. The structure of CNN can be seen briefly in figure 1. YOLOv3's small object (such as bird droppings, leaves) detection accuracy is poorer, but YOLOv3 can avoid background error to produce false positives, and you can see the global image of context information. On the other hand, YOLOv3 detection speed is faster, so the first step in the system when handling a large number of scene image using YOLOv3 can be more quickly to facilitate further processing, greatly improve the recognition rate of the system. For the problem that small target is difficult to detect, Mask R-CNN with high recognition accuracy is adopted, ResNet and FPN are used as backbones. Before entering the network, the image is cropped and then scaled, which increases the time of network training and prediction, but greatly improves the accuracy and improves the accuracy of the system.

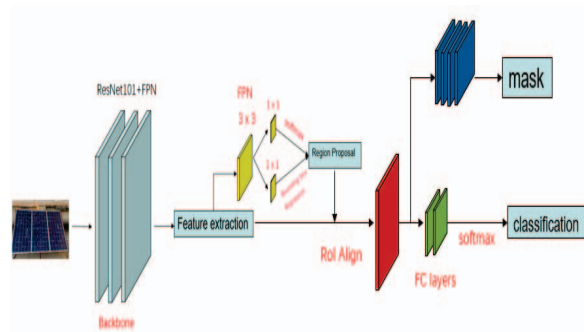


Figure 1: Overview of method .

2.1 YOLOv3

Unlike region-based approaches, you only look once (YOLO) [17] is a state-of-the-art, real-time object detection system. YOLOv2 [18] overcomes the relatively high localization error and low recall (measure of how good is the localization of all objects). In 2018, YOLOv3 is released and is characterized by a higher accuracy and replaces softmax function with logistic regression and threshold. The main improvement made with YOLOv3 is the use of new CNN feature extractor named Darknet-53 [19]. It also uses

3x3 and 1x1 convolutional layers the same as ResNet [20]. Experimental result demonstrates that YOLOv3 own both high accuracy and better speed, compare with ResNet-152, it has less floating point operations but the same accuracy at a double speed.

2.2 Mask R-CNN

The Mask-RCNN model advance the Faster-RCNN [21] model for semantic segmentation, object localization, and object instance segmentation of natural images. Mask R-CNN can be regarded as a combination of ResNet and FPN [22] in general. In principle, the backbone network could be any convolutional neural network designed for images analysis, such as ResNet-50 or ResNet-101; However, it has been shown that using a feature pyramid network (FPN) based on a network such as ResNet-50 or ResNet-101 as the Mask-RCNN backbone gives gains in both accuracy and speed.

2.2.1 ResNet

In order to overcome the problem of the disappearance of the gradient of the deep network, Kaiming He et al. proposed a deep residual network ResNet, which improved the model recognition effect at the same time. Residual block is a kernel technique in ResNet. Residual block create a shortcut in the input of each two convolutional layers and perform an element-wise addition to the outputs to maintain the influence of the original input and avoid gradients disappear.

2.2.2 Feature Pyramid Network(FPN)

In order to better represent the object on multiple scales, the feature pyramid network was proposed to extend the backbone network, which is also effective for the detection of small objects. The top-level features of FPN architecture were merged with the underlying features by up-sampling, with each layer independently predicting feature maps.

3 PV Abnormal Shading Detect System

In this section, we established a detection system for PV module abnormal shading, including PV panel detection, abnormal shading location and recognition, as shown in the figure 2.



Figure 2: The flow of system.

3.1 PV Panel Detection

Due to the interference of environmental factors in the acquired field photovoltaic panel image (aerial shooting,

etc.), the extraction of the photovoltaic panel first and then the detection will not only greatly improve the accuracy of foreign body recognition, but also be crucial to the subsequent positioning of the fault photovoltaic panel. In this part, we first use labeling to label the field images manually. In order to improve the robustness of the model, we only train the yolov3 model with the labeled PV panel images at this stage, then extract and process the photovoltaic panel images collected in the field, at last sets the portions of the detected panels as RoI (region of interest) and proceeds to the next step to locate the abnormal shading.

3.2 Abnormal Shading Location

This section continues to use the YOLOv3 algorithm to locate the abnormal shading on the photovoltaic panels, on account of leaves, bird drops have different shapes and angles, we use 7 different anchor boxes, apply the seven anchor boxes to each grid center to detect abnormal shading, then capture the abnormal shading image to the final phase for character recognition.

3.3 Shading Classification

Compared with yolov3, Mask R-CNN has a slower detection speed, but has a better effect on feature extraction of edges and a higher mAP detection of targets, which is convenient for subsequent pixel detection and classification. Because of the different shapes and sizes of shading, we use the deeper ResNet101-FCN as the backbone. Full convolution network (FCN) is used for labeling process of the ROI, then we use Mask R-CNN to classify the abnormal shading. In the first two parts, we used YOLOv3 to extract the photovoltaic panel and locate the foreign body to increase the proportion of the object in the image. Under the matting of the first two steps, we can improve the recall rate of the character recognition and improve the accuracy of abnormal shading recognition.

4 Experiment Results and Analysis

4.1 Dataset

The quality of dataset has a great impact on the performance of the model. In this experiment, we created a detailed scheme to collect pictures and expand the dataset, which can improve the accuracy of the algorithm and effectively prevent the over-fitting and make the model more generalized.

4.1.1 Image Collection

In this experiment, an experimental platform was built to simulate the arrangement of PV panels in a photovoltaic plant. Three photovoltaic panels are used for experiment. In this experiment, pictures of leaves of different shapes (oval, strip) and bird drops of different shapes (speck, strip) attached to PV panels were collected. The canon30 camera was used for shooting in a fixed position, and an image was collected every 1 minute between 11:00 AM and 13:00 PM for a month as a dataset. There were 5,400 pictures in the dataset, including 1,000 pictures containing only leaves, 1,000 pictures containing only bird droppings and 3,400 pictures containing both leaves and bird

droppings. When training the model, the training-set and verification-set were respectively based on the ratio of 8:2.

4.1.2 Dataset Augmentation

In order to prevent over-fitting and further improve the performance of the model, we expanded the dataset: to simulate the photovoltaic panel with dust accumulation, we added gaussian noise to some images; we got small samples through randomly sample patches, which improved detection accuracy of smaller objects.

4.2 System Evaluate and Analysis

First, in the photovoltaic panel extraction step, we used the expanded dataset to train the YOLOv3 model, which only contained labels with PV panels, to improve the robustness of the model. As for result, the precision (P) and recall (R) rates are applied to evaluate the object detection performance of YOLOv3:

$$P = \frac{TP}{TP + FP}$$

$$R = \frac{TP}{TP + FN}$$

where TP is the number of cases that are positive and detected positive, FP is the number of cases that are negative but detected positive, and FN is the number of cases that are positive but detected negative. The results of 1000 test images indicate that the overall precision rates and recall rates are 96.03% and 95.77%, as showed from table1.

Parameter	Leaves	Bird-Drops	Over All
Precision Rate	97.31	94.75	96.03
Recall Rate	97.01	94.53	95.77

Table 1: Precision and Recall Rate

For the abnormal shading classification stage, in order to explore a faster and more concise method, we compared three different backbone with Mask R-CNN for the experiment. The experimental results are shown in the table2: ResNet101, ResNet50-FPN, and ResNet101-FPN.

Backbone	mAP	IoU
ResNet-101	84.59	88.86
ResNet-50+FPN	87.45	91.22
ResNet-101+FPN	92.17	93.82

Table 2: Mask Results

In order to verify the effectiveness of the system, we directly trained the pictures of Mask R-CNN. Through the comparison of recall rates, it can be seen that the recognition accuracy after PV panel extraction and shading location is higher, as shown in table 3, proving that this model can well comfort the needs of photovoltaic field.

5 Conclusion

In this paper, our system is designed to detect the abnormal shading of PV module with high precision. The experimental results indicate that our system classification accuracy rate is above 94%. It is convenient for the follow-up study of the surface shading of PV modules.

Model Type	Mask R-CNN		System		Improvement
	Leaves	Bird-Drops	Leaves	Bird-Drops	
Oval/Strip	85.43	80.71	95.41	93.23	22.4
Strip/Speck	82.81	82.11	93.97	93.86	22.91

Table 3: System Performance Comparison

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